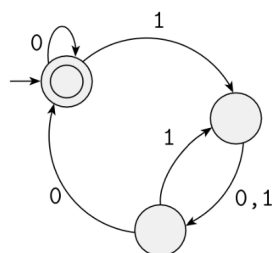


Assignment 8

Q1

Answer all parts for the following DFA M and give reasons for your answers.



- | | |
|---|---|
| a. Is $\langle M, 0100 \rangle \in A_{DFA}$? | d. Is $\langle M, 0100 \rangle \in A_{REX}$? |
| b. Is $\langle M, 011 \rangle \in A_{DFA}$? | e. Is $\langle M \rangle \in E_{DFA}$? |
| c. Is $\langle M \rangle \in A_{DFA}$? | f. Is $\langle M, M \rangle \in EQ_{DFA}$? |

A1

- (a) Yes. The DFA M accepts 0100.
- (b) No. M doesn't accept 011.
- (c) No. This input has only a single component and thus is not of the correct form.
- (d) No. The first component is not a regular expression and so the input is not of the correct form.
- (e) No. M 's language isn't empty.
- (f) Yes. M accepts the same language as itself.

Q2

Consider the problem of determining whether a DFA and a regular expression are equivalent. Express this problem as a language and show that it is decidable.

A2

Equivalence problem for DFAs

- The **equivalence problem** for DFAs is that of determining whether or not two DFAs recognize the same language.

Let $EQ_{DFA} = \{\langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$

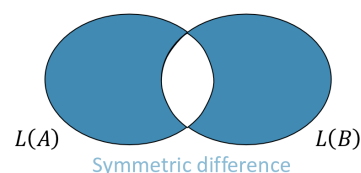
Is it decidable?

Theorem: EQ_{DFA} is decidable

Proof: Give TM D_{EQ-DFA} that decides EQ_{DFA} .

D_{EQ-DFA} = "On input $\langle A, B \rangle$ [IDEA: Make DFA C that accepts w where A and B disagree.]

- Construct DFA C where $L(C) = (L(A) \cap \overline{L(B)}) \cup (\overline{L(A)} \cap L(B))$.
- Run D_{E-DFA} on $\langle C \rangle$.
- Accept if D_{E-DFA} accepts.
Reject if D_{E-DFA} rejects."



Assume the regular expression is R and the DFA is M , we describe the algorithm as follows,

1. Convert the regular expression R to an equivalent NFA N .
2. Convert the NFA N to an equivalent DFA N' .
3. Run TM $D_{\text{EQ-DFA}}$ on input $\langle M, N' \rangle$.
4. Accept if $D_{\text{EQ-DFA}}$ accepts, reject if not.

Q3

Let $ALL_{\text{DFA}} = \{ \langle A \rangle \mid A \text{ is a DFA and } L(A) = \Sigma^* \}$. Show that ALL_{DFA} is decidable.

A3

Emptiness Problem for DFAs

- The **emptiness problem** for DFAs is that of determining whether or not a DFA accepts or not any string at all.

Let $E_{\text{DFA}} = \{ \langle B \rangle \mid B \text{ is a DFA and } L(B) = \emptyset \}$

Is this decidable?

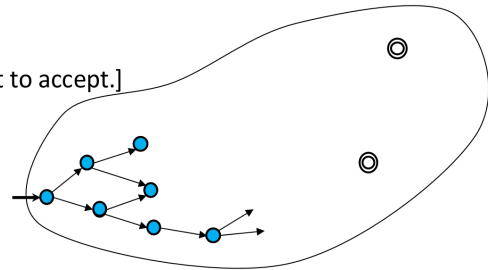
Can we try the DFA on all possible strings?

Theorem: E_{DFA} is decidable

Proof: Give TM $D_{\text{E-DFA}}$ that decides E_{DFA} .

$D_{\text{E-DFA}}$ = "On input $\langle B \rangle$ [IDEA: Check for a path from start to accept.]

1. **Mark** start state.
2. Repeat until no new state is marked:
Mark every state that has an incoming arrow from a previously marked state.
3. *Accept* if no accept state is marked.
Reject if some accept state is marked."



Give TM $D_{\text{ALL-DFA}}$ that decides ALL_{DFA} .

$D_{\text{ALL-DFA}}$ = " On input $\langle A \rangle$

1. Construct DFA C where $L(C) = \overline{L(A)}$.
2. Run $D_{\text{E-DFA}}$ on $\langle C \rangle$.
3. Accept if $D_{\text{E-DFA}}$ accepts, reject if not. "

Q4

Let $A\epsilon_{\text{CFG}} = \{ \langle G \rangle \mid G \text{ is a CFG that generates } \epsilon \}$. Show that $A\epsilon_{\text{CFG}}$ is decidable.

A4

Version 1:

$D_{A\epsilon\text{-CFG}}$ = " On input $\langle G \rangle$

1. Mark all the occurrences of terminal ϵ in G .
2. Repeat until no new variables are marked:
Mark all occurrences of variable A if $A \rightarrow B_1 B_2 \dots B_k$ is a rule and all B_i were already marked.
3. Accept if the start variable is marked, reject if not. "

Version 2:

$D_{A_\varepsilon\text{-CFG}} =$ " On input $\langle G \rangle$

1. Convert G into CNF G'
2. Check if G' contains the production rule $E \rightarrow \varepsilon$
3. If it contains, accept; otherwise, reject. "

Q5

Let $E_{\text{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$. Show that $\overline{E_{\text{TM}}}$, the complement of E_{TM} , is Turing-recognizable.

A5

Let $s_1, s_2, \dots$ be a list of all strings in Σ^* . The following TM recognizes $\overline{E_{\text{TM}}}$.

"On input $\langle M \rangle$, where M is a TM:

1. Repeat the following for $i = 1, 2, 3, \dots$
2. Run M for i steps on each input, $s_1, s_2, \dots, s_i$.
3. If M has accepted any of these, *accept*. Otherwise, continue."

Q6

Prove that EQ_{DFA} is decidable by testing the two DFAs on all strings up to a certain size. Calculate a size that works.

A6

Suppose that we have two DFAs A and B . Set new DFA C that

$$L(C) = (L(A) \cap \overline{L(B)}) \cup (\overline{L(A)} \cap L(B)).$$

Apparently, $Q_C = \{(p, q) \mid p \in Q_1, q \in Q_2\}$.

For C , $|Q_C| = |Q_1||Q_2|$. The length of the path from start state to any other state is at most $|Q_C| - 1$. Then by taking all paths of length $|Q_C| - 1$, it either can finally reach the states in F_C or can check if $F_C = \emptyset$.

Thus, by running C on strings whose length is up to $|Q_1||Q_2| - 1$, we can get the results of whether the two DFAs accept the same language.